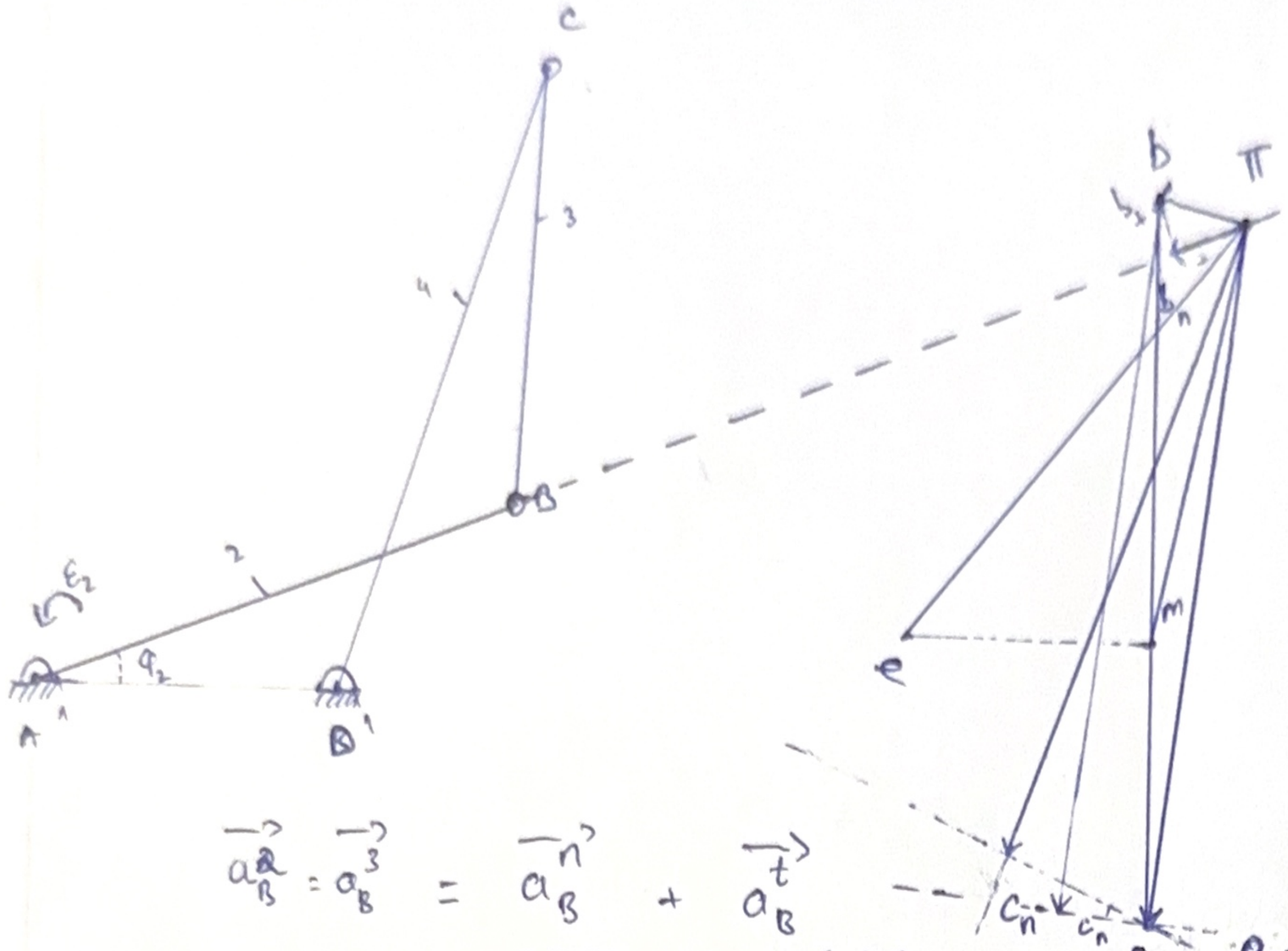


$$\vec{a}_B^{\,2} = \vec{a}_B^{\,3} = \vec{a}_B^{\,n} + \vec{a}_B^{\,t}$$

$$\begin{cases} \parallel BA \\ B \to A \\ l_{AB}.\omega_2^2 = 125 \ m/s^2 \end{cases} \quad \begin{cases} \perp BA \\ \varepsilon_2 \\ l_{AB}.\varepsilon_2 = 100 \ m/s^2 \end{cases}$$

$$a_B = \sqrt{a_B^{n^2} + a_B^{t^2}} \approx 160,078 \ m/s^2$$

1) Phương trình gia tốc tại C

$$\vec{a}_c^{\,n} + \vec{a}_c^{\,t} = \vec{a}_B^{\,n} + \vec{a}_B^{\,t} + \vec{a}_{CB}^{\,n} + \vec{a}_{BC}^{\,t}$$

$$\begin{cases} \parallel CD \\ C \to D \\ \omega_4^2.l_{CD} = 1209,84 \ m/s^2 \end{cases} \quad \begin{cases} \perp CD \\ ? \\ \varepsilon_4.l_{CD} \end{cases} \quad \begin{cases} \parallel AB \\ B \to A \\ 125 \ m/s^2 \end{cases} \quad \begin{cases} \perp AB \\ \varepsilon_2 \\ 100 \ m/s^2 \end{cases} \quad \begin{cases} \parallel BC \\ C \to B \\ \omega_3^2.l_{BC} = 1296 \ m/s^2 \end{cases} \quad \begin{cases} \perp BC \\ ? \\ ? \end{cases}$$

Chọn gốc gia tốc π, $\vec{\pi b}$ dùng để biểu diễn $\vec{a}_B$

Cho $b_n = 6,25$mm $\Rightarrow \mu_a = \dfrac{a_B^n}{|b_n|} = \dfrac{20 \ m/s^2}{mm}$

$a_c = |\pi c|.\mu_a \approx 63,4.20 \approx 1268 \ m/s^2 \Rightarrow 1268^2 = 1209,84^2 + (\varepsilon_4 - l_{CD})^2$

$\Rightarrow \varepsilon_4 \approx 64,3 \ rad/s^2$ (cùng chiều kim đồng hồ)

$a_{bc}^t = |cc_n'|.\mu_a \approx 9,9.20 \approx 198 \ m/s^2 \Rightarrow \varepsilon_3 = \dfrac{a_{bc}^t}{l_{BC}} \approx 49,5 \ rad/s^2$ (cùng chiều kim đồng hồ)

+) $m \in bc$ với $\dfrac{BM}{BC} = \dfrac{2,5}{4} = \dfrac{5}{8} \Rightarrow bm \approx 41$ mm.

$a_M = |\pi m|.\mu_a \approx 38,75.20 \approx 775 \ m/s^2$

+ $\Delta BCE \sim \Delta bcq \Rightarrow \dfrac{ME}{BC} = \dfrac{me}{Bc} \Rightarrow me = bc.\dfrac{ME}{BC} = 24,6$ mm.

$a_E = |\pi_e|.\mu_a \approx 50.20 \approx 1000 \ m/s^2$